

Homework — 1.5.2 Moral and Ethical Issues

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.5.2 lesson.

Part A — This week's topic (~14 marks)

1. Match each term to its definition: write the correct letter in the **Match** column. Two letters are not used.
[3] (AO1)

Term	Match
Automated decision making	
Algorithmic bias	
"Black box" system	
Offshoring	
Monopoly	
Digital divide	

Letter	Definition
A	A computer system makes or recommends a decision with little or no human involvement.
B	A system performs worse for some groups of people because its training data under-represented them.
C	A system whose internal reasoning cannot easily be inspected or explained.
D	Moving jobs or operations to another country where costs are lower.
E	One company dominates a market, weakening competition.
F	The gap between those who have access to digital technology and skills and those who do not.
G	Software whose source code is freely available to view, use and modify.
H	Using someone's creative work without the copyright holder's permission.

2. A police facial-recognition system misidentifies people from some ethnic groups far more often than others. The jumbled statements below describe how this bias arose. Number them **1–5** in the correct order. [2] (AO2)

Step	Order (1–5)
The deployed system misidentifies people from the under-represented groups far more often.	
Engineers train the model on this dataset.	
A dataset of faces is collected that under-represents some ethnic groups.	
The police force puts the system into use for identification.	
The model learns facial features mainly from the over-represented groups.	

3. A hospital introduces an AI system that reads scans and flags likely tumours for a radiologist to check.

(a) State **one** benefit of using automated decision making here. [1] (AO2)

(b) Give **one** ethical concern of relying on this system to make the initial decision. [1] (AO2)

4. Complete the paragraph using words from the word bank. Each item may be used at most once; **two** items are not needed. [2] (AO1)

When a single company dominates a market it holds a _____, which can lead to higher prices and vendor _____ because customers depend on one supplier's proprietary product. Open-source software counters this because its _____ is freely available to view, use and modify, so users can switch supplier or maintain the software themselves instead of being tied to a single _____.

Word bank: monopoly · lock-in · source code · vendor · copyright · patent

5. A council moves all of its services — benefits claims, bin collections, parking permits — to "online only". **Discuss** the moral and social impact of this decision on the different groups affected, and reach a justified conclusion. [5] (AO3)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.5.1) A hacker breaks into a company server *intending* to steal trade secrets to sell to a rival. State the Act and the specific **offence** committed. [2] (AO1)

7. (1.3.3) State the purpose of the **TCP** layer and the purpose of the **IP** layer in the TCP/IP stack. [2]
(AO1)

8. (1.1.2) State **one** reason a **multi-core** processor can complete a workload faster than a single-core processor of the same clock speed. [2] (AO1)
